

T VARATHARAJAN

3D Engine Developer —WebGL • WebGPU • Graphics Programming • CAD Systems

Chennai, Tamil Nadu, India · +91 9659309280 · ciazvanish2000@gmail.com

linkedin.com/in/t-varatharajan · github.com/Varatharajan1808 · varatharajan - portfolio

Professional Summary

3D Engine Developer with 2.5+ years of experience building browser-based graphics engines and CAD applications using WebGL, WebGPU, and TypeScript. Experienced in rendering pipelines, graphics engine architecture, GLSL/WGSL shader development, Physically Based Rendering (PBR), scene graph systems, GLTF asset pipelines, and GPU optimization. Passionate about developing scalable, high-performance real-time rendering solutions for the web.

Technical Skills

- Programming Languages : JavaScript , TypeScript, GLSL, WGSL, HTML5, CSS3
- Graphics APIs : WebGL, WebGPU
- Graphics Programming : Rendering Pipelines, Shader Development, Scene Graphs, Real-Time Rendering, Physically Based Rendering (PBR), Lighting Models, HDRI Environment Mapping, GPU Optimization
- CAD & Geometry : Browser-Based CAD Systems, Geometry Processing, Raycasting, Object Picking, Interactive Editing
- Asset Pipelines : GLTF 2.0, GLB, PBR Materials, Texture Optimization
- Tools & Build Systems : Git, GitHub, Webpack, Vite, NPM, VS Code, Chrome DevTools, Jira

Professional Experience

3D Engine Developer

Aalai Immersive Tech Pvt Ltd — Chennai, Tamil Nadu, India

May 2026 – Present

- Developing browser-based CAD systems using a custom WebGPU-based rendering engine.
- Implementing CAD entities, geometry processing, and interactive modeling workflows.
- Building reusable rendering components and enhancing WebGPU-based CAD capabilities.

Software Developer (Web-Based 3D Graphics– WebGL/WebGPU)

Hexr Factory Immersive Tech Pvt Ltd

Dec 2023 – Apr 2026

- Developed a custom WebGL/WebGPU-based rendering engine for browser-based CAD and real-time 3D graphics applications.
- Implemented geometry systems, scene graph architecture, rendering pipelines, and Physically Based Rendering (PBR).
- Developed GLSL/WGSL shader systems, real-time lighting, camera controls, and GPU-optimized rendering techniques.
- Built and optimized GLTF/GLB asset pipelines for efficient real-time rendering.
- Improved rendering performance through efficient buffer management, render-loop optimization, and GPU resource management.
- Collaborated with cross-functional teams to develop scalable engine architecture and ensure cross-browser compatibility.

Key Projects

Personal Projects

1. Vara3D Graphics Engine

GitHub: github.com/Varatharajan1808/Vara3D-Graphics-Engine

Live Demo: <https://varatha3deditor.vercel.app>

- Developed a browser-based 3D graphics engine and visual editor from scratch using TypeScript and WebGL.
 - Designed and implemented rendering pipelines, scene graph architecture, GLSL shader systems, and Physically Based Rendering (PBR).
 - Built a complete GLTF/GLB asset pipeline supporting meshes, materials, textures, normal maps, and hierarchical scene structures.
 - Implemented editor features including scene hierarchy, transform gizmos, material editing, texture management, and real-time rendering.
 - Optimized rendering performance using frustum culling, efficient buffer management, and GPU resource optimization.
-

2. LuminaSphere

GitHub: github.com/Varatharajan1808/luminasphere-webgl

Live Demo: <https://luminasphere-webgl.vercel.app>

- Developed an interactive WebGL-based 3D orb using TypeScript, GLSL, and gl-matrix.
 - Implemented shader-driven animations, dynamic color palettes, and responsive mouse/touch interactions.
 - Built custom icosahedron geometry and adaptive camera behavior for desktop and mobile devices.
 - Designed a lightweight WebGL application showcasing real-time graphics programming and shader development.
-

3. Portfolio Website

GitHub: github.com/Varatharajan1808/Varatharajan_Portfolio

Live Demo: <https://varatharajanportfolio.vercel.app>

- Built a personal portfolio using TypeScript and a custom component architecture with reusable UI components and state management..
 - Implemented reusable components, state management, responsive layouts, and interactive UI animations.
-

Education

Bachelor of Science in Mathematics

Jul2017–Apr2020

Kamaraj College, Manonmaniam Sundaranar University – Thoothukudi, Tamil Nadu, India

MBA in HumanResourceManagement (HRM)

Dec2021–May2023

Alagappa University – Thoothukudi, Tamil Nadu, India

Certifications

- Full Stack Web Development – QSpiders Institute, Chennai

Apr 2023